Andrew Zeng

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EDUCATION

Northeastern University **Boston, MA**
B.S. in Electrical Engineering, Minor in Computer Science *May 2028*
Honors: NU College of Engineering Merit Scholarship, Global Scholars Program
Activities: NU Lunabotics, Students for the Exploration and Development of Space, Club Golf, Club Spikeball
Relevant Coursework: Embedded Design, Discrete Structures, Fundamentals of CS I & II, Computing for Engineers

SKILLS

Languages: Python, C, C++, MATLAB, JavaScript/TypeScript
Hardware & Embedded: PCB Design, Circuit Debugging, Microcontrollers, I²C, Soldering, Crimping, Oscilloscopes, DMMs
Tools & Systems: KiCad, AutoCAD, RF/DSP, PLC Ladder Logic, SolidWorks, Git, Linux, GitHub

EXPERIENCE

Electrical Engineering & Test Engineering Co-op **New York, NY**
Opentrons Labworks Inc. *July 2026 – Present*
- Executed subsystem-level testing and validation of robotic hardware throughout prototyping, development, and production
- Created a Hall-effect measuring system spanning design and firmware integration, analyzing tip-offset to ~0.03mm accuracy
- Built and debugged circuits, PCB's and fixtures using oscilloscopes, DMM, and power supply, with

Electrical Engineering Intern (OEM Parts Division) **Pittsburgh, PA**
Westinghouse Electric Company *April 2026 – July 2026*
- Designed mechanical and electrical hardware for breaker product lines servicing AP1000 and operating PWR nuclear plants
- Built automated stress-testing device cycling circuit breakers through continuous linear actuation, executing 1000+ test cycles
- Performed root-cause failure analysis on circuit breakers, identifying defects and failures, revising refurbishment procedures

Research Intern **Ann Arbor, MI**
University of Michigan *May 2025 – August 2025*
- Fine-tuned and integrated YOLO models in Python on imagery to build a perception pipeline for autonomous-driving systems
- Designed and programmed an autonomous car integrating onboard sensing, perception algorithms, and closed loop control
- Presented model results and perception findings into university lecture content in collaboration with faculty and research staff

PROJECTS

Autonomous Swarm Robots (Personal) *Aug 2026 - Present*
- Secured $10,000 in funding from OpenAI to develop autonomous general-purpose swarm robots using collaborative SLAM
- Architected the complete robot platform, selecting sensors, compute hardware, motor drivers, power electronics, and chassis
- Simulated collaborative SLAM to evaluate multi-robot localization, mapping, and information sharing across the swarm

Hall-Effect Measurement System (Opentrons) *July 2026 - Present*
- Design Hall-effect sensing system to measure pipette-tip offset across varying tip geometries for liquid-handling lab robots
- Iterate magnet shape, sensor selection, placement, and orientation through in-fixture testing to characterize sensor response
- Write Python test scripts interfacing with robot firmware and issuing Unix-level commands to automate sensor data collection

Automated Circuit Breaker Stress Testing Device (Westinghouse Electric) *May 2026 - July 2026*
- Modeled frame and switch-flipping linkage, using FEA to validate against 100+ lb breaker weight and operational shock loads
- Developed integration of linear actuator, control system, limit switches, load cell, and power distribution for automated cycling
- Programmed PLC ladder logic for cycle counting, end-of-travel interlocks, and fault handling across automated test sequence

FMCW Radar (Personal) *Feb 2026 - Present*
- Developed an FMCW radar detection and tracking system, processing raw I/Q data through a custom Python DSP pipeline
- Implemented range-Doppler FFTs, clutter suppression, and detection thresholds to estimate target range and radial velocity
- Developed frame-to-frame target association for tracking moving objects and visualizing detections in real time

Custom Bus PCB (Northeastern Lunabotics Team) *Jan 2026 - Mar 2026*
- Designed a custom I²C backbone PCB in KiCad to coordinate sensor boards across the sensor and power stack of the rover
- Created full KiCad schematics with signal routing, connectors, and interface logic for reliable inter-board communication
- Implemented a power-management architecture distributing regulated power to sensor components across the board